OPENNESS OF VERSALITY FOR THE FUNCTOR F

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6451-6455 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 6457-6497, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_artin.json (pb=2236, stmt 6451-6455, proof 6457-6497).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GHM: sources/stacks-artin/pinned-original-source.tex, source lines 6497–6497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GI9: sources/stacks-artin/pinned-original-source.tex, source lines 6330–6340. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH8: sources/stacks-artin/pinned-original-source.tex, source lines 5138–5153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0APN: sources/stacks-artin/pinned-original-source.tex, source lines 6497–6497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0GH8 the functor F satisfies openness of versality.*

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We have to show the following. Given a scheme V locally of finite type over S, given $\xi \in F(V)$, and given a finite type point $v_0 \in V$ such that ξ is versal at v_0, after replacing V by an open neighbourhood of v_0 we have that ξ is versal at every finite type point of V. Write $\xi = (Z, u', \hat{x})$.

First case: $v_0 \notin Z$. Then we can first replace V by $V \setminus Z$. Hence we see that $\xi = (\emptyset, u', \emptyset)$ and the morphism $u' : V \to X'$ is versal at v_0. By More on Morphisms of Spaces, Lemma 0APN this means that $u' : V \to X'$ is smooth at v_0. Since the set of a points where a morphism is smooth is open, we can after shrinking V assume u' is smooth. Then the same lemma tells us that ξ is versal at every point as desired.

Second case: $v_0 \in Z$. Write $W = \operatorname{colim} W_n$ as in Formal Spaces, Lemma 0GHM. By Lemma 0GI9 we may assume $\hat{x} : V_{/Z} \to W$ is a smooth morphism of formal algebraic spaces. It follows immediately that $\xi = (Z, u', \hat{x})$ is versal at all finite type points of Z. Let $V' \to V$, $\hat{x}'$, and x' witness the compatibility between u' and $\hat{x}$. We see that $\hat{x}' : V'_{/Z} \to X'_{/T'}$ is smooth as a base change of $\hat{x}$. Since $\hat{x}'$ is the completion of $x' : V' \to X'$ this implies that $x' : V' \to X'$ is smooth at all points of $(V' \to V)^{-1}(Z) = |x'|^{-1}(T') \subset |V'|$ by the already used More on Morphisms of Spaces, Lemma 0APN. Since the set of smooth points of a morphism is open, we see that the closed set of points $B \subset |V'|$ where x' is not smooth does not meet $(V' \to V)^{-1}(Z)$. Since $V' \to V$ is proper and hence closed, we see that $(V' \to V)(B) \subset V$ is a closed subset not meeting Z. Hence after shrinking V we may assume $B = \emptyset$, i.e., x' is smooth. By the discussion in the previous paragraph this exactly means that ξ is versal at all finite type points of V not contained in Z and the proof is complete. $\square$